



ARTIFICIAL INTELLIGENCE

Class XI

STUDENT HANDBOOK



Subject Code:843

UNIT 4: Introduction to Capstone Project

Title: Introduction to Capstone Project	Approach: Hands on, Team Discussion, Web search, Case studies
Summary: In this unit, students will be able to understand the concept of Capstone project and its importance as a final project of an academic program. They will understand that all the knowledge gathered will be integrated in the form of a project known as Capstone Project. The chapter will help the students understand how real-life simple day to day problems or situations can be handled and solved using a methodology known as Design Thinking and how to leverage AI in addressing these issues. The students will also learn to associate the issues with Sustainable Development Goals. They will be able to exchange their points of view based on experiences and discuss potential solutions to the problem.	
Learning Objectives: 1. Understand the meaning of the Capstone Project and its goals. 2. Understand how problems can be identified, decomposed and solved using Design Thinking Methodology. 3. Learn the steps of Design Thinking and apply for solving simple issues. 4. Learn to create Empathy maps. 5. Understand the importance of 5W1H in Design Thinking and Capstone Project development. 6. Relate the importance of Sustainable Development Goals and how these issues can be aligned with Capstone Project.	
Key Concepts: 1. Design Thinking 2. Empathy Map 3. Sustainable Development Goals 4. Capstone Project	
Learning Outcomes: By the end of this unit, students will be able to 1. Decompose any problem using the 5W1H method 2. Apply Design thinking methodology 3. Create empathy maps 4. Align problems to SDGS 5. Apply all the learnings in solving real world problems 6. Comfortably express their solution to a problem in non-technical words	
Pre-requisites: Basic knowledge about problem solving, issues around them, good communication skills to express their views about any issue, basic knowledge about AI, scientific temper.	

Introduction to Capstone Project

A capstone project is a project where students must research a topic, deeply understand the subject matter and integrate all their knowledge gathered about a particular subject to develop a solution to a problem.

The AI Capstone project will give learners an opportunity to implement the AI skills as a final step in the learning path to showcase their expertise in the subject matter.

While doing a Capstone project, students will learn more about problems which they can solve to improve lives and make the world a better place. Also, they learn many important skills including:

- How to work as a team member?
- How to clearly identify an issue? and how it is affecting (the user)?
- How to brainstorm solutions and select the best one?
- How to decide which type of AI may be useful for the proposed solution?
- How to ethically gather and use data to train a computer to help solve the issue?
- How to test the prototype with users and use their feedback to improve the solution?
- How to pitch their solution to people who will be able to help them take action?

Some Examples for Capstone Project are given below

1. Stock Prices Predictor
2. Develop A Sentiment Analyzer
3. Movie Ticket Price Predictor
4. Students Results Predictor
5. Human Activity Recognition using Smartphone Dataset
6. Classifying humans and animals in a photo

The list of Examples is huge as every year students are coming up with new ideas and innovations for new projects.

Asking the right question before starting the AI Capstone Project

Before Starting a Project, one should analyze whether a problem has a pattern associated with it. If there is no pattern, then the problem cannot be solved with AI technology. It is fundamental that this question, **“Is there a Pattern?”** is asked before deciding to embark on an AI development journey.

There are various AI techniques which can be applied to check for patterns and solve the problems. Predictive Analysis helps in identifying the patterns by asking different questions which fall into different approaches. Some of the questions are given below:



- 1) Which category? (Classification)
- 2) How much or how many? (Regression)
- 3) Which group? (Clustering)
- 4) Is this unusual? (Anomaly Detection)
- 5) Which option should be taken? (Recommendation)

Now, once these questions are answered, we can decide that the problem can be solved using AI. Complex problems cannot be dealt with as such. Then we try to simplify the complex problem to analyze it. This is known as Problem Decomposition.

Problem decomposition steps

1. Understand the problem and then restate the problem in your own words.
 - Know what the desired inputs and outputs are?
 - Ask questions for clarification (in class these questions might be to your instructor, but most of the time they will be asking either yourself or your collaborators)
2. Break the problem down into a few large pieces.
 - Write these down, either on paper or as comments in a file.
3. Break complicated pieces down into smaller pieces.
 - Keep breaking down until all of the pieces are small.
4. Code one small piece at a time.
 - Think about how to implement it? and write the code/query.
 - Test it on its own and fix problems, if any.



Activity

1. Many people are facing serious issues with traffic jams during office hours which may be caused due to multiple reasons. Considering this as a major issue to resolve, write down the problem decomposition steps to make it simple to solve.

First Step → Understand the Problem and Restate it.

Public is facing difficulty due to the problem of traffic jams especially during the busy office hours.

Second Step → Break the problem down into a few large pieces.

Split the traffic issue separately for each reason

- a. Bad Roads
- b. Accidents
- c. Office Hours
- d. Rash Driving
- e. Inappropriate Signals
- f. Over Crowded Area

Third Step → Break complicated pieces down into smaller pieces.

From the above sub steps (reasons), if any reason is of complicated type, we can split it into more simpler units. For example, Rash Driving can be further split into:

- a. Over Speed
- b. New Drivers with less experience
- c. Careless Driving

Fourth Step → Take each issue at a time and try to find the solution.

Critical and Creative Thinking

Definition: Critical thinking is the ability to analyze a situation and make a judgment based on facts and data. Creative thinking on the other hand refers to the ability to come up with new ideas or rather solutions.

So, these two skills are very important to raise questions against problems, gather and assess relevant information, brainstorm ideas and formulate well-reasoned conclusions and solutions. Creative thinking and critical thinking are necessary skills for equipping individuals to be the social change makers, leaders and innovators. Complex problems can be easily solved using critical and creative thinking which has a systematic approach to address any problem which has to be solved. This approach is known as Design Thinking.

Introduction to Design Thinking

Definition: Design thinking is a non-linear, iterative process that expert teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test.

Design Thinking framework

Design Thinking is a design methodology that provides a solution-based approach to solving problems. It's extremely useful in tackling complex problems that are ill-defined or unknown.

The five stages of Design Thinking are as follows: Empathize, Define, Ideate, Prototype, and Test.

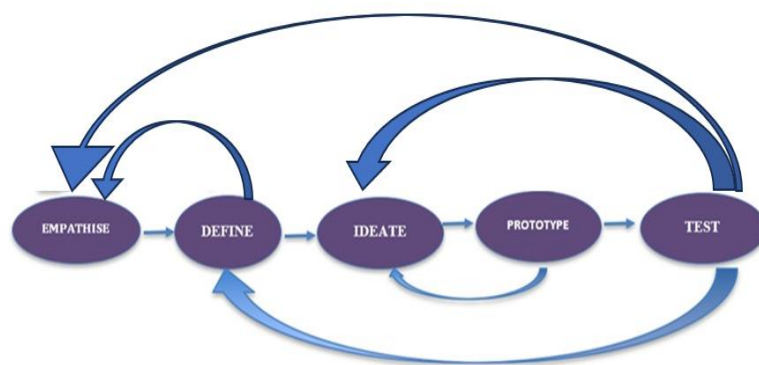


Figure 1: Stages of Design Thinking Framework

i) Empathize

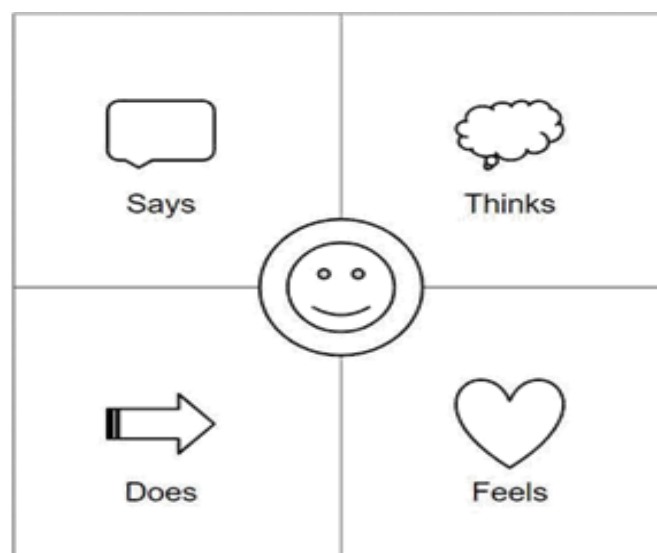
Design thinking begins with empathy. This requires doing away with any preconceived notions and immersing oneself in the context of the problem for better understanding. In simple words, through empathy, one is able to put oneself in other people's shoes and connect with how they might be feeling about their problem, circumstance, or situation.

There is a challenge one needs to solve. How does one approach it? Empathy starts from here. As a designer of the solution to a challenge, one should always understand the problem from the end-user perspective. This is done by observation, interaction or by imagination.

Designers are expected to interact with customers / users very frequently to gather detailed facts about the problems and user's expectations. A detailed analysis of these facts leads to approaching the problem in the best possible way.

Empathy Map

An extremely useful tool for understanding the users' needs and gaining a deeper insight into the problem at hand is the empathy map. It also helps in deepening that understanding, gaining insight into the user's behavior.



To create a “persona” or profile for the user, you can use the empathy map activity to create a realistic general representation of the user or users. Personas can include details about a user's education, lifestyle, interests, values, goals, needs, thoughts, desires, attitudes, and actions.

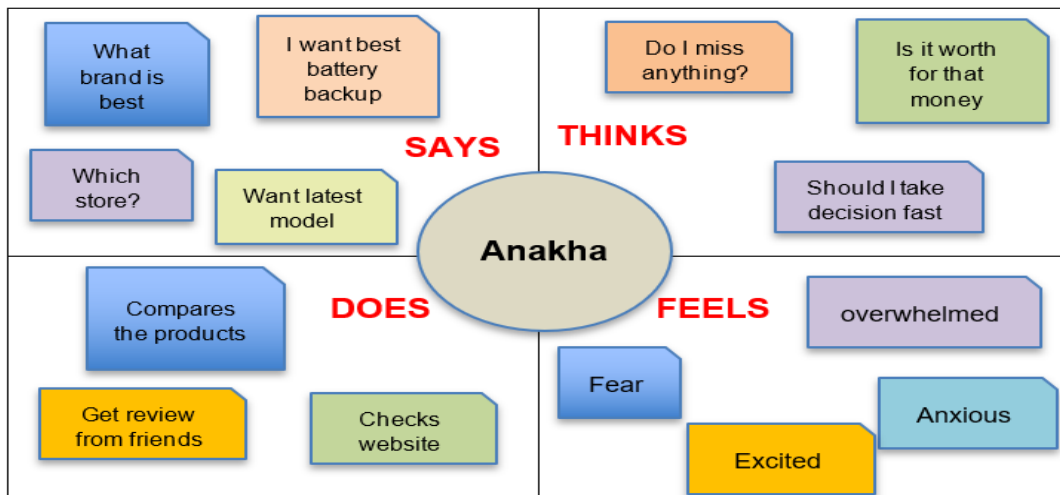
An Empathy Map is divided into 4 quadrants. —Says, Thinks, Does, Feels

- Says – This quadrant contains whatever the user says aloud
- Thinks – This quadrant depicts the thoughts which user have about the problem
- Does – This quadrant depicts the actions of user
- Feels – This quadrant displays the emotional status of the user

Creating Empathy Map

Draw the Empathy map grid as shown above. Write down the respective observations in corresponding quadrants. These observations can be written directly or can paste a sticky note with the observations in the quadrants. Let us understand through an example. Anakha is having a desktop computer. She is planning to buy one new Laptop for her educational purpose. Let us create an Empathy map for Anakha.

Empathy Map for Anakha



Online tool to create Empathy Map

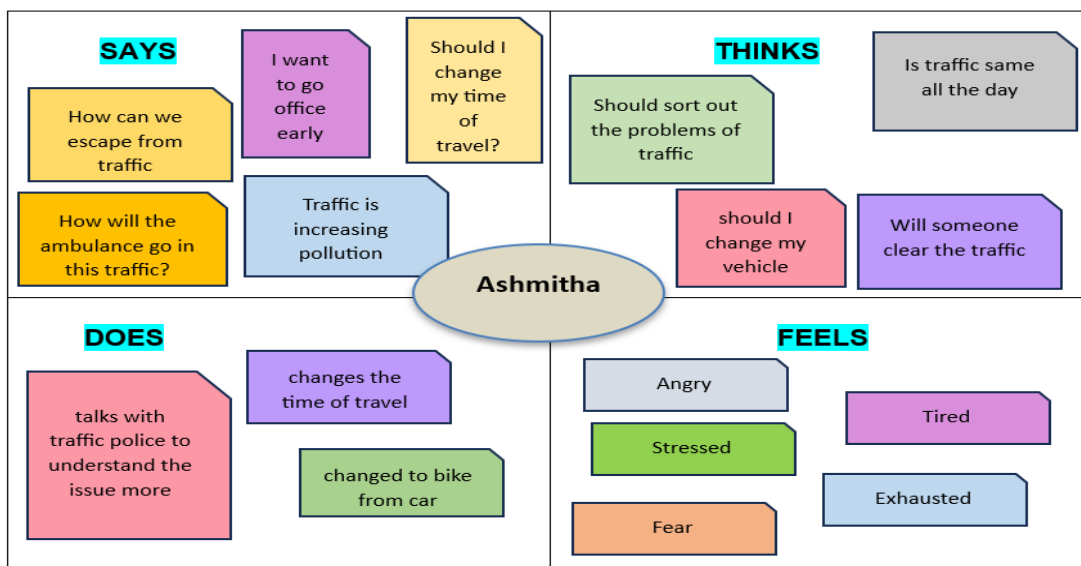
<https://online.visual-paradigm.com/diagrams/features/empathy-map-template/>

Reference: <https://www.ibm.com/design/thinking/page/toolkit/activity/empathy-map>



Activity

1. Ashmitha daily drives to her office and back. The office is hardly 30 minutes' drive from her home. However, due to traffic jams it takes more than 1 hour. Ashmitha is hoping for a solution to this traffic issue. Prepare an Empathy map related to Ashmitha.



ii) Define

In the Define stage, information collected during Empathize is used to draw insights and is instrumental in stating the problem that needs to be solved. It's an opportunity for the design thinker to define the challenge or to write the problem statement in a human-centered manner with a focus on the unmet needs of the users.

In order to extract / gather relevant facts and information from users/customers, it is recommended to use this simple and reliable method of questioning: the **5W1H method**—i.e. asking questions starting with Who? What? When? Where? Why? and How?

For Example: For the traffic issue in a city, we could define the problem in detail after getting the answers of 5W1H questions. Sample questions of the problem is given below

Who ?	What ?	Where ?
• Who are the sufferers? • Who is responsible? • Who can control it? • Who all are involved in it? • Who needs immediate help? • Who questions it? • Who identifies it?	• What can be done? • What should be done? • What is the issue? • What causes the issue? • What steps are necessary? • What to be avoided? • What may be the reason?	• Where is the problem seen? • Where else it can occur? • Where to start the solution? • Where to concentrate more? • Where are the issues severe? • Where can we ignore issue? • Where is solution impossible
When ?	Why ?	How ?
• When does traffic occurs? • When can we start clearing the traffic? • When is it more problematic • When to think about a quick solution? • When is traffic ignorable?	• Why traffic occurs? • Why it becomes a problem? • Why should we find a solution to it? • Why to clear traffic in that city? • Why can't we ignore traffic?	• How will we find a solution? • How to deal with the traffic? • How to manage the users affected by traffic? • How to solve without creating more issues? • How to identify the traffic?



Activity

1. Considering Ashmitha's traffic issue answer the following questions given below in order to help define the problem.

a. Who is the sufferer in the Problem?

b. What is the Problem?

c. Where does it happen?

d. When can it happen usually?

e. Why does it happen?

f. How can it be solved?

iii) Ideate

By now the problem is obvious and it is time to brainstorm ways and methods to solve it. At this stage, numerous ideas are generated as a part of the problem-solving exercise. In short, ideation is all about idea generation. During brainstorming, one should not be concerned if the generated ideas are possible, feasible, or even viable. The only task of the thinkers is to think of as many ideas as possible for them. It requires "going wide" mentally in terms of concepts and outcomes. There are many brainstorming tools that can be used during this stage. Here is an overview of the most essential ideation techniques employed to generate numerous ideas.

Brainstorm

During a Brainstorming session, students leverage the synergy of the group to generate new innovative ideas by building on others' ideas. Participants should be able to discuss their ideas freely without fear of criticism. A large number of ideas are collected so that different options are available for solving the challenge.

Brain dump

Brain dump is very similar to Brainstorm; however, it's done individually. It allows the concerned person to open the mind and let the thoughts be released and captured onto a piece of paper. The participants write down their ideas onto paper or post-it notes and share their ideas later with the larger group.

Brain writing

Brain writing is also very similar to a Brainstorm session and is known as 'individual brainstorming'. At times only the most confident of team members share their ideas while the introverts keep the ideas to themselves. Brainwriting gives introverted people time to write them down instead of sharing their thoughts out loud with the group. The participants write down their ideas on paper and, after a few minutes, pass on their own piece of paper to another participant who then elaborates on the first person's ideas and so forth. In this way all participants pass their papers on to someone else and the process continues. After about 15 minutes, the papers are collected and posted for instant discussion.

During brainstorming, one should not be concerned if the generated ideas are possible, feasible, or even viable. The only task of the thinkers is to think of as many ideas as possible for them. It requires "going wide" mentally in terms of concepts and outcomes. It is all about creativity and imagination; all types of ideas are encouraged, whether stupid or wise – it hardly matters as long as the solution is imagined.

After everyone shares their ideas, specific measures are applied to evaluate the ideas without being judgmental or critical to narrow the list. It may so happen that the solution comes from the unlikeliest of ideas. So, at this point focus is on quantity over quality of ideas. The most feasible ideas are chosen for further exploration. Storyboarding, or making a visual mock-up of an idea, can also be useful during ideation.



Activity

1. For the traffic problems in a city, brainstorm and generate different ideas as a team.

iv) Prototype

The prototype stage involves creating a model designed to solve consumers' problems which is tested in the next stage of the process. Creating a prototype is not a detailed process. It may include a developing simple drawing, poster, group role-playing, homemade “gadget, or a 3D printed product.” The prototypes must be quick and easy to develop and cheap. Therefore, prototypes are visualized as rudimentary forms of what a final product is expected to look like. Prototyping is intended to answer questions that get you closer to your final solution. Prototypes, though quick and simple to make, bring out useful feedback from users. Prototypes can be made with everyday materials also.



Activity

1. Through a poster or simple drawing depict a prototype for any one solution for the traffic issue.

v) Test

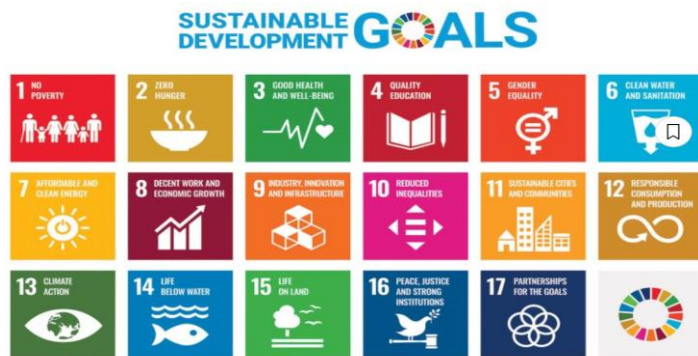
One of the most important parts of the design thinking process is to test the prototypes with the end users. This step is often seen going parallel to prototyping. During testing, the designers receive feedback about the prototype(s), and get another opportunity to interact and empathize with the people they are finding solutions for. Testing focuses on what can be learned about the user and the problem, as well as the potential solution.



Activity

1. Show the prototype made by you to your parents/teachers/relatives who usually get stuck in traffic issues. Collect the feedback from them and understand whether your prototype will work or not. Write down the views your users shared and the conclusion you arrived at.

Sustainable Development Goals



Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs. The global community is witnessing a transition towards a sustainable future. To achieve a sustainable future, 17 goals have been announced by the

United Nations which are termed as the Sustainable Development Goals.

Projects undertaken all over the world play a crucial role in fulfilling this task by aligning their strategies and vision with the sustainable development goals and targets.

Students can select any problem based on SDG to find an AI solution for it.

For example:

SDG 2 was adopted to achieve “Zero Hunger”. That means it is essential to recognize that the successful transformation of our current food systems (shift to a sustainable food system) should tackle the main global challenges in a sustainable, healthy, and inclusive manner. Healthy food habits, food Supply to all areas, consumption of food as per the production, Seasonal Foods etc. are the areas where students can concentrate for identifying problems and try to derive simple AI based solution.

Some other use cases of the SDGs are given below

- Good health and well being
- Gender equality
- Renewable energy
- Combating hunger and poverty
- Protecting resources
- Promoting responsible consumption and production
- Providing access to education
- Sustainable and safe cities and communities



Activity

1. Write down a few problems which you notice from any five SDGs which can be taken up as a Capstone project.

Project Abstract Creation Using Design Thinking Framework

We can now choose a problem (capstone project) and check how it can be detailed using Design Thinking Framework.

Format for Project Abstract Creation

1. Project Name:
2. Team Members Name:
3. Problem Selection –Important issue which you think can be solved using AI by doing survey or interview and which is aligned to any of the SDGs.
4. Users Affected by the Problem:
5. Empathise – create Empathy Map
6. Define the Problem – Write down the 5W1H questions and write the problem statement
7. Ideate – Brainstorm and write down the ideas
8. Prototype- Create prototype by simple drawing, poster, group role-playing, homemade “gadget, or a 3D printed product.”
9. Test (optional at this stage)

Now let us see an example

- 1. Project Name:** {Suitable name related to your problem and your creativity}
- 2. Team Members Name:** { Students can fill by themselves}
- 3. Problem Selection:** After doing a survey amongst the school students who are in Class X, it was observed that most of the students are confused about the subjects that they have to choose in future.

Details about the issue:

While taking admission to plus-two courses, students are confused with subject selection. To provide quality education students should be provided with the subjects in which they have aptitude. A system to analyze their interests, choose subjects from the available options and choose the institutions which provide those subjects are a big problem. After seeing the issue, it was aligned to SDG-4, Quality Education.

4. Users:

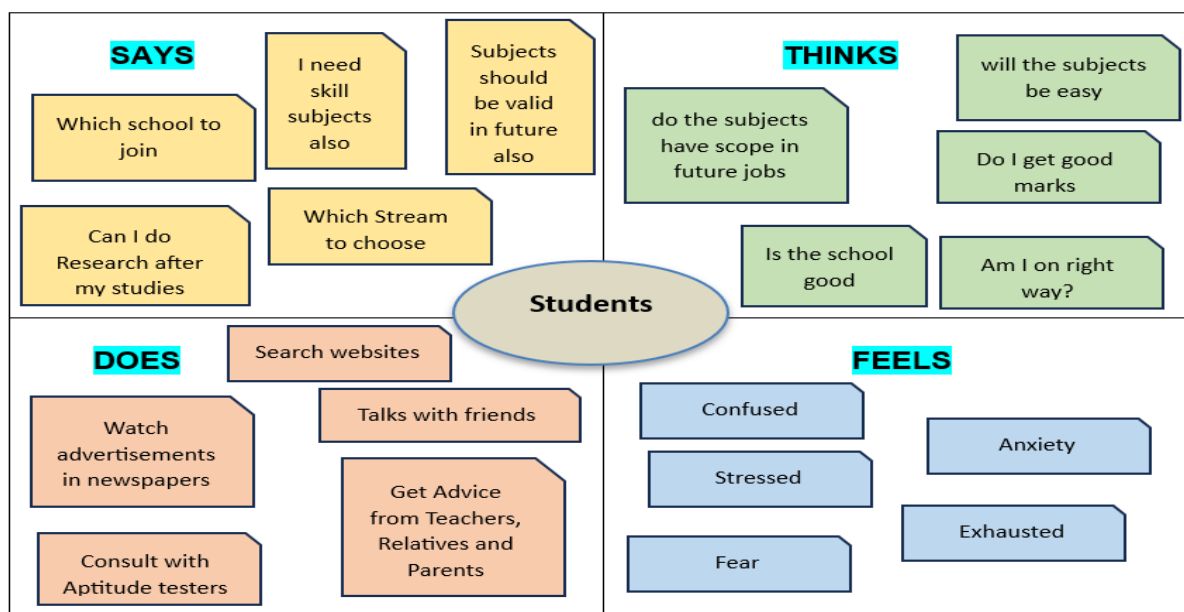
All the students who pass class X face the issue of selecting an appropriate subject for their higher studies as it is their turning point to move on according to their interest.

5. Empathise:

Two categories of students can be met and interviewed.

- First—Those students who are studying in plus-two (to know how difficult it was to choose a subject and a school)
- Second—Those students who are going to take admission (can know about their anxieties and expectations)

After the discussion with students, team members can sit together and finalize the points they understood from the users(students). They can write each in a chit of paper, classify it into the categories SAYS, THINKS, DOES & FEELS. Now they can create an Empathy Map. A sample empathy map is given below.



6. Define:

After Empathetically understanding the students, the problem may be defined. Get the Answers of 5W1H Questions. Write down the Problem Statement.

5W1H Questions & Answers

Who?

1. Who are the users?

Students seeking admission in plus-two course.

2. Who else can be the users?

Parents/well-wishers who seek admission for their wards.

3. What do you know about the users?

Users are students who are interested in taking plus-two admissions. Students have well defined future plans. Now they have to select the appropriate subject as per their plan.

What?

1. What is the Problem?

Not able to select the subject of their choice.

2. What else are the Problems?

They can't choose the school which provides these subjects.

Where?

1. Where is the problem/ In which situation does the problem arises?

All the places where admission takes place & all the place where students wish to take admission the problem persists.

When?

1. When does the problem occur?

Problem occurs during the time of admission.

Why?

1. Why does the problem occur?

Problem occurs due to the lack of knowledge of subjects available as part of the curriculum.

2. Why should it be solved?

Only if students pursue the course, they have aptitude, a well-developed society can be built in future.

How?

1. How can we solve it?

Through a system which will analyze a student's aptitude and suggests the subject to take. After checking the database, the system can suggest the schools giving the subject combinations also.

2. How will the solution improve the situation?

The solution could help students in choosing the appropriate subjects and could give effective education.

PROBLEM STATEMENT

Our students have the problem of not being able to choose the subjects of their interest while taking admission to their plus-two course and it can be solved by developing a system which will analyze students' aptitude and suggest the subject to take.

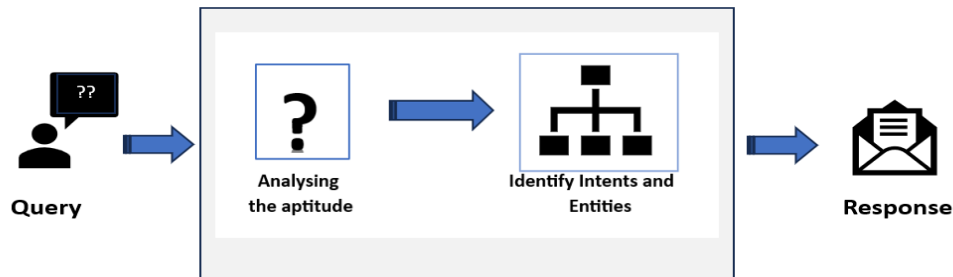
7. Ideate:

As a team, students can discuss and put forward different ideas to implement the solution.

1. Create an application to input the interest and get suggestions on subjects.
2. Create a chatbot to converse with.
3. Make a robot to discuss with and give advice.
4. Create a mobile application where through different sources, queries can be given and the application will give the result.

8. Prototype:

Select any one idea and prepare the prototype of the solution - Here we select the idea of creating a chatbot to give the solution.



Prototype of Working of a chatbot

Details of intents and entities can be drawn separately to display the prototype in detail. (Leaving it to the creativity of students)

EXERCISES

A. Multiple Choice Questions:

1. Which is NOT correct about the Capstone project?
 - a. It is a project where students must research independently
 - b. Students can integrate all their knowledge to develop a solution to the problem
 - c. It is the final project of an academic program.
 - d. Students are made to look at real world situations, exchange their point of view and discuss potential solutions to the problem.
2. Which is the fundamental question which is asked before deciding to embark on an AI development journey?
 - a. Should we do the project?
 - b. Is there a solution?
 - c. Can we leave the problem?
 - d. Is there a pattern?
3. The question, 'which option should be taken?' comes under which category?
 - a. Classification
 - b. Regression
 - c. Anomaly detection
 - d. Recommendation
4. Which stage in Design Thinking is missing (Prototype, Ideate, Test, Define)?
 - a. Evaluation
 - b. Empathise
 - c. Evolution
 - d. Enrichment
5. Real computational tasks are complicated. To accomplish them, you need to _____ before coding.
 - a. Test the problem causing variables
 - b. Ask for finance help from stakeholders
 - c. Break down the problem into smaller units
 - d. write it in a paper and read it to your partners

6. Which one is not a part of the 5W1H method?
 - a. Who
 - b. While
 - c. Where
 - d. When
7. _____ is the process of generating ideas and solutions through sessions such as sketching, brainstorming etc.
 - a. Define
 - b. Empathise
 - c. Ideate
 - d. Prototype
8. Which one among the following is known as individual brainstorming?
 - a. Brainwriting
 - b. Brain dump
 - c. Brainstorm
 - d. Brain explore
9. _____ is a useful process during ideation.
 - a. Storyboarding
 - b. Empathy map
 - c. Prototyping
 - d. Data collection
10. The _____ stage involves creating a model designed to solve consumer's problems in Design Thinking.
 - a. Modeling
 - b. Prototype
 - c. Training
 - d. Deployment
11. _____ Analysis helps in identifying the patterns by asking different questions which fall into different approaches.
 - a. Predictive
 - b. Prescriptive
 - c. Probability
 - d. Prototype

B. Short Answer Questions

1. What do you mean by a capstone project?
2. Name some the skills which students will acquire by doing capstone project.
3. Give some examples of use case of SDG's.
4. Students are promoted to develop critical and creative thinking. Why?
5. Define Design Thinking.

C. Long Answer Questions

1. Write down the steps of Problem decomposition.
2. Explain briefly the different stages of Design thinking.
3. Explain the different types of Ideation techniques.
4. Mr. Muthu Kumar is facing a serious issue of high electricity bill every month. He is trying a lot to reduce it. Considering his issue is worth solving, draw an empathy map related to Muthu Kumar.

D. Case Study

1. Based on SDG-7, Affordable and clean energy, formulate a Problem and write down the Design thinking steps associated with it.